Random Walk

$$\begin{cases} X_t = X_{t-1} + W_t \qquad W_t \overset{iid}{\sim} N(0,1) \\ X_0 = 0 \end{cases}$$

$$X_t = W_1 + \cdots + W_t$$

Claim: for fixed t,

$$X_t \sim N(0, t)$$

We already showed: $E[X_t] = 0$

$$Var(X_t) = Var(W_1 + \cdots + W_t)$$

due to independence of W_t's.

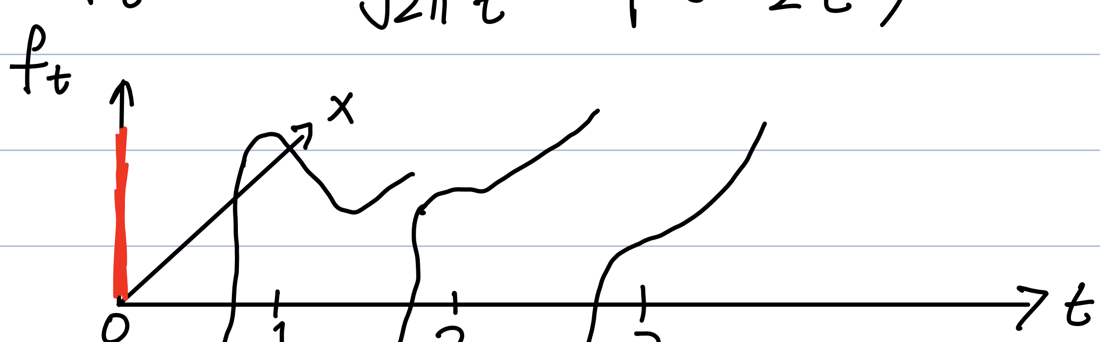
$$= Var(W_1) + \cdots + Var(W_t)$$

$$= \overbrace{1 + \cdots + 1}^{t \text{ of "1"s}}$$

$$= t$$

$\Rightarrow$ PDF at time t

$$f_t(x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right)$$



Remark: We know now the RW x_t has marginal distribution $N(0,t)$.
Make sure to distinguish this from another setup:
$Y_1 \sim N(0,1)$, $Y_2 \sim N(0,2)$, ...
where Y_i's are independent

In particular, for RW x_t, if x_{t-1} is known, it helps a lot to predict $x_t = x_{t-1} + w_t$

However, this is NOT the case for Y_t's.

- Joint distribution (Multivariate distribution)

Random vector (multivariate RV)
$\vec{x} = (x_1, \cdots, x_n)$

RVs
may be dependent on each other

One can think of a time series as such a long vector.

Joint CDF*: $F(c_1, \cdots, c_n) = P(X_1 \le c_1, X_2 \le c_2, \cdots, X_n \le c_n)$

Joint PDF*: $\dfrac{\partial^n}{\partial c_1 \cdots \partial c_n} F(c_1, \cdots, c_n) = f(c_1, \cdots, c_n)$

Remark:

Question: can we statistically learn the joint distribution of the random vector consisting of the entire time series $(X_1, X_2, \cdots, X_n)$?

↳ Motivating Question:

Suppose X is a RV, and we observe a realized value $X = x$.

Q: Can we learn about the distribution of X?

Answer: very challenging:

We only observe one realization of the random vector $(X_1, \cdots, X_n)$.

In addition, even from a theoretical perspective, joint distr. of a very high. dim. is also challenging to study.

Def. $\{X_t\}$: a time series

The mean function is defined as:

$$\mu(t) = E[X_t] = \int x f_t(x)\, dx$$

Rem: Mean function captures the mean trend of the time series.

E.g. Moving Average:

$$X_t = \frac{1}{3}(w_{t-1} + w_t + w_{t+1})$$

w_t: white noise.

$$\mu(t) = E[X_t]$$

$$= E\left[\frac{1}{3}(w_{t-1} + w_t + w_{t+1})\right]$$

$$= \frac{1}{3}E[w_{t-1}] + \frac{1}{3}E[w_t] + \frac{1}{3}E[w_{t+1}]$$

$= 0$

E.g.: Random walk with drift.

$$\begin{cases} x_t = \underbrace{\delta}_{\text{a non-random constant}} + x_{t-1} + w_t \\ x_0 = 0 \end{cases}$$

Recall: $x_t = \delta t + w_1 + \cdots + w_t$

$$\mu(t) = E[x_t] = E[\delta t + w_1 + \cdots + w_t]$$

$$= \underbrace{E[\delta t]}_{\text{NOT random}} + \underbrace{E[w_1] + \cdots + E[w_t]}_{0}$$

$$= \delta t$$

$\Rightarrow \mu(t)$: a linear function of time.

Question:

"Variance function"

$$Var(x_t) = Var(\delta t + \underline{w_1 + \cdots + w_t})$$

?

Important Fact:
For a random variable X, & a non-random number c. $Var(X+c) = Var(X)$.

$$= Var(w_1 + \cdots + w_t) \overset{\text{known}}{=} t$$

E.g. Signal plus noise:

$X_t = S_t + U_t$ where

S_t: non-random signal

U_t: random noise s.t. $E[U_t] = 0$.

$$\mu(t) = E[X_t] = E[S_t + U_t]$$
$$= E[S_t] + E[U_t]$$
$$= S_t + 0 = S_t$$

Def. $\{X_t\}$: a time series.
The covariance function
(autocovariance function)
is defined as:

$\gamma(s,t) = Cov(X_s, X_t)$ $\left(= Cov(X_t, X_s)\right)$

$$= E\left[(X_s - \mu(s)) \cdot (X_t - \mu(t))\right]$$

Remark :

- $\gamma(s,t) = \gamma(t,s)$ (symmetric)

- $\gamma(t,t) = Var(X_t)$

- Typically speaking for a time series.
  when $\gamma(s,t)$ decays slowly when $|s-t| \uparrow$,
  $\Rightarrow$ time series looks "smooth".
  when $\gamma(s,t)$ decays fast when $|s-t| \uparrow$,
  $\Rightarrow$ time series look "rough".

Exercise (Ex 10 in Rmarkdown homework notebook)

$\rightarrow \gamma(s,t) = E[X_s X_t] - \mu(s)\mu(t)$

E.g. w_t: white noise. $Var(w_t) = \sigma_w^2$

$$\gamma(s,t) = Cov(w_s, w_t)$$

$$= \begin{cases} \sigma_w^2 & \text{if } s=t \\ 0 & \text{if } s \neq t \end{cases}$$

Fact: $\mathrm{Cov}(X, Y)$ behaves like a product between X & Y.

RVs : X, Y, Z, U.

$$\mathrm{Cov}(X+Y, Z+U)$$

$$(X+Y)\cdot(Z+U)$$

$$= \mathrm{Cov}(X,Z) + \mathrm{Cov}(Y,Z) + \mathrm{Cov}(X,U) + \mathrm{Cov}(Y,U)$$

$$= XZ + YZ + XU + YU$$

c: non-random constant

$$\mathrm{Cov}(cX, Y)$$

$$(cX)\cdot Y$$

$$= c\,\mathrm{Cov}(X,Y)$$

$$= c\,(X\cdot Y)$$

A formal statement of this property: (bilinearity)

$$U = a_1 X_1 + \cdots + a_m X_m$$

$$V = b_1 Y_1 + \cdots + b_r Y_r$$

X_i, Y_i's: RVs

a_i, b_i's: non-random

$$Cov(U, V) = \sum_{j=1}^{m} \sum_{k=1}^{n} a_j b_k Cov(X_j, Y_k)$$